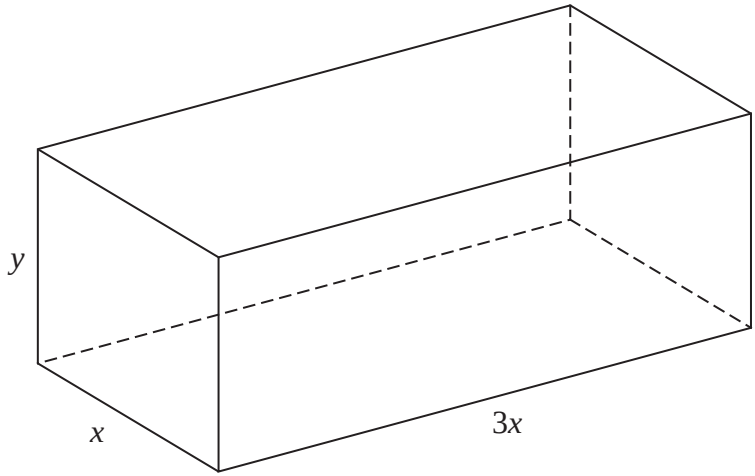



**Figure 1**

Figure 1 shows an open-topped container used for holding water.

The container is in the shape of a cuboid and is made of sheet metal.

The base of the container is a rectangle $3x$ metres by $x$ metres.

The height of the container is $y$ metres as shown in Figure 1.

Given that the capacity of the container is $120\,\text{m}^3$

(a) show that the area $A\,\text{m}^2$ of the sheet metal used to make the container is given by

$$A = Px^2 + \frac{Q}{x}$$

where $P$ and $Q$ are positive constants to be found.

**(4)**

(b) Use calculus to find the value of $x$ for which $A$ has a stationary value, giving your answer to 3 significant figures.

**(4)**

(c) Find $\frac{\mathrm{d}^2A}{\mathrm{d}x^2}$ and hence show that the value of $x$ found in part (b) gives the minimum value of $A$.

**(2)**